

Advanced (Habanero)

Q1. Solve the inequality $2x + 5 > 11$.

- (A) $x > 3$
- (B) $x < 3$
- (C) $x > -3$
- (D) $x < -3$
- (E) $x = 3$

Q2. Graph the solution to $x - 2 < 7$ on a number line.

- (A) $(-\infty, 9)$
- (B) $(-\infty, 5)$
- (C) $(-\infty, 9]$
- (D) $[5, \infty)$
- (E) $(-9, \infty)$

Q3. Solve the inequality $x/4 + 2 > 5$.

- (A) $x > 12$
- (B) $x < 12$
- (C) $x > -6$
- (D) $x < -6$
- (E) $x = 12$

Q4. Write the solution to $2x + 5 < 11$ in interval notation.

- (A) $(-\infty, 3)$
- (B) $(-\infty, 3]$
- (C) $[3, \infty)$
- (D) $(-3, \infty)$
- (E) $(-\infty, -3)$

Q5. Solve the compound inequality $-3 < 2x + 1 < 5$.

- (A) $-2 < x < 2$
- (B) $-1 < x < 2$
- (C) $-2 < x < 1$
- (D) $-1 < x < 1$
- (E) $-2 < x < 3$

Q6. Solve the inequality $|x - 2| > 3$.

- (A) $x < -1$ or $x > 5$
- (B) $x < 1$ or $x > 3$
- (C) $x < -3$ or $x > 1$
- (D) $x < -5$ or $x > -1$
- (E) $x < 5$ or $x > 1$

Q7. Solve the inequality $x^2 - 4x - 3 > 0$.

- (A) $x < -1$ or $x > 5$
- (B) $x < 1$ or $x > 3$
- (C) $x < -3$ or $x > 1$
- (D) $x < -1$ or $x > 3$
- (E) $x > -1$ or $x < 3$

Q8. Write the solution to $x^2 + 2x - 6 > 0$ in interval notation.

- (A) $(-\infty, -3) \cup (1, \infty)$
- (B) $(-\infty, -1) \cup (3, \infty)$
- (C) $(-3, 1)$
- (D) $(-1, 3)$
- (E) $(-3, -1) \cup (1, 3)$

Open Ended Questions (FRQ)

Q9. Solve the inequality $3x - 2 > 2x + 5$, and graph the solution on a number line. Write the solution in interval notation.

Q10. Solve the inequality $x/2 + 1 > 3$, and write the solution in interval notation.

Chef's Tips

- Make sure students show their work and explain their reasoning.
- Remind students to check their answers.
- Encourage students to use graphing to visualize the solutions.